

4.0 now appearing in most new devices, it now offers the similar range and bandwidth as that of Wi-Fi but with a lot less power consumption.

Since Bluetooth is backwards compatible, your old accessories can be used with new devices without much issues, but you'd not be able to reap the benefits of the new features that each revision brings with it. Here is a simple table highlighting the differences between different Bluetooth versions.

Wi-Fi

Wi-Fi or IEEE 802.11 is an older protocol for data transfer but consumed much more power and offered higher range and bandwidth. Wi-Fi too has undergone multiple revisions with multiple improvements in each revision that led to a much efficient system for data transfer.



Wi-Fi can connect to a host of devices to interconnect and access the internet. This method of connection can require a mediator as in most centralised/decentralised/hybrid systems, however, there also exist methods to enable ad-hoc networks (simple closed group, even one-to-one) like Wi-Fi Direct.

The current revision of Wi-Fi is 802.11ac, however, the average user won't be able to saturate anything above 802.11n. The current internet services simply don't offer services that'd need more bandwidth than 802.11n. Then again since it uses a different frequency band there is less interference compared to existing radio technologies. It is the most common and widely used protocol, so anything above that is a luxury. The good thing about this is that it is comparatively power efficient and thus better suited for mobile application and each successive revision will be better.

Here is a simple table to compare existing Wi-Fi protocols:

	Wi-Fi(a)	Wi-Fi(b)	Wi-Fi(g)	WiMAX
Standard	802.11a	802.11b	802.11g	802.16
Frequency (GHz)	5	2.4	2.4	2.66
Speed Mbps	54	11	54	80
Range	50m	100m	100m	50km
Advantages	Speed	Low cost	Speed	Speed, Range
Disadvantages	Cost	Speed	Cost, Range	Cost
Radio technology	OFDM	Direct sequence spread spectrum	OFDM (64 - channels)	OFDM (256 - channels)
Mobility	In development	In development	In development	Mobile WiMAX (802.16e)
Primary application	Wireless LAN	Wireless LAN	Wireless LAN	Broadband wireless access